

Phase 4: Feature Engineering, Train/Test Evaluation, and Model Improvement

In Phase 3, you built a Simple Linear Regression model, evaluated it, and wrote a reflection identifying what it got wrong. You were not done. You were setting up for this.

Phase 4 is your response to your own critique. You will engineer new features, apply a proper train/test split to get an honest estimate of model performance, and test whether your improvements actually help. At the end, you will produce a structured summary that will become the backbone of your final presentation.

The goal of this phase is to:

- Understand why Phase 3 evaluation was optimistic and fix it
- Engineer at least one new feature and justify it with evidence
- Retrain and evaluate your models on a proper train/test split
- Compare all models in a single unified table
- Produce a presentation-ready summary of your full project arc

Submission Requirements

Submit the following:

- Jupyter Notebook (.ipynb)
- All dataset files used, including any supplementary data you added
- A short README section at the top of your notebook including:

Your README must contain:

- Original research question from Phase 1
- Your target variable and all features used across Phase 3 and Phase 4
- Final dataset shape used for modeling (rows x columns)
- One sentence describing what changed between Phase 3 and Phase 4 and why

Part 1: Why Phase 3 Was Optimistic

Before writing any new code, you need to understand what was wrong with your Phase 3 evaluation.

In Phase 3, you trained your model on the full dataset and then evaluated it on that same full dataset. This is called in-sample evaluation. The model has already seen every data point it is being tested on. Of course it performs reasonably well. You are not measuring how well it predicts new data. You are measuring how well it memorized the data it was trained on.

This is the fundamental problem that train/test splits solve. You hold out a portion of your data before training, the model never sees it, and you evaluate on that held-out set. That score is an honest estimate of how your model would perform on data it has never encountered before.

Think of it like a practice exam versus the real exam. A student who studies only the practice exam questions and then takes the same exam again will score perfectly. That score tells you nothing about whether they actually learned the material. The real exam has new questions. Train/test split is the real exam.

You must write 3 to 5 sentences answering the following in your notebook:

- What was the evaluation setup in Phase 3, and why is it problematic?
- What does a train/test split fix, in your own words?
- Do you expect your Phase 3 RMSE to go up or down when evaluated on held-out data, and why?

Part 2: Apply a Train/Test Split

Split your dataset before doing anything else in this phase. This is not optional and the order matters. If you engineer features before splitting, your split is compromised.

Use the following structure:

```
from sklearn.model_selection import train_test_split

X = df[['feature']]    # your Phase 3 feature
y = df['target']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Print the shape of your training and test sets. Confirm that your split produced approximately an 80/20 division.

Now retrain your Phase 3 Simple Linear Regression on the training set only and evaluate it on the test set:

```
from sklearn.linear_model import LinearRegression
import numpy as np

model = LinearRegression()
```

```
model.fit(X_train, y_train)

y_pred_test = model.predict(X_test)
rmse_test = np.sqrt(np.mean((y_test - y_pred_test) ** 2))
print('Test RMSE:', rmse_test)
```

Compare this test RMSE to your Phase 3 RMSE. Write 2 to 3 sentences answering: did your model perform better or worse on held-out data? Was the difference large or small? What does that tell you about how well your Phase 3 model had actually generalized?

Part 3: Feature Engineering

Now you improve your model. Feature engineering is the process of creating new input variables from your existing data to give your model more useful information.

You must implement at least two of the following transformations and justify each one with evidence from your data:

3a: Log or Power Transformation

If your feature or target has a skewed distribution, a log or square root transformation can linearize the relationship and reduce the influence of outliers.

```
df['log_feature'] = np.log1p(df['feature'])    # log1p handles zeros
safely
```

You must produce a before and after scatter plot showing the relationship between your transformed feature and your target. Write 2 to 3 sentences explaining whether the transformation helped and how you know.

Only apply a log transformation if it is justified by your data. If your feature is already linear with your target, transforming it may make things worse. Your job is to test, not to assume.

3b: Interaction Term

An interaction term captures the combined effect of two features. It is the product of two variables and tests whether their relationship with the target is multiplicative rather than additive.

```
df['interaction'] = df['feature_1'] * df['feature_2']
```

Justify your interaction term in 1 to 2 sentences. Why do you believe these two variables might have a combined effect on your target? What would it mean if the interaction term improves your model?

3c: Binning a Continuous Variable

Sometimes a continuous variable has a nonlinear relationship with your target that is better captured by grouping it into categories. For example, age might matter differently at different life stages rather than scaling linearly.

```
df['feature_binned'] = pd.cut(df['feature'], bins=4, labels=False)
```

If you use binning, you must justify why you believe the relationship between your feature and target is not linear. Cite your Phase 3 residual plot as evidence if applicable.

3d: Supplementary Data (Optional but encouraged)

If your original dataset did not give you enough signal, you may bring in a supplementary dataset and merge it. This must be a real, citable data source.

If you do this, you must:

- Identify the source and explain why it is relevant
- Show the merge operation and confirm the resulting dataset shape
- Explain what new information the supplementary data provides that your original data did not

Part 4: Retrain and Evaluate

Now retrain your model using your engineered features. Use the same train/test split from Part 2. Do not re-split.

```
X_engineered = df[['feature_1', 'log_feature', 'interaction']] # your actual features
```

```
X_train_eng = X_engineered.loc[X_train.index]
```

```
X_test_eng = X_engineered.loc[X_test.index]
```

```
model_eng = LinearRegression()
```

```
model_eng.fit(X_train_eng, y_train)
```

```
y_pred_eng = model_eng.predict(X_test_eng)
```

```
rmse_eng = np.sqrt(np.mean((y_test - y_pred_eng) ** 2))
```

```
print('Engineered Model Test RMSE:', rmse_eng)
```

Produce a residual plot for your engineered model using the test set residuals only:

```
residuals_eng = y_test - y_pred_eng
plt.scatter(y_pred_eng, residuals_eng)
plt.axhline(0, color='red')
plt.xlabel('Predicted Values')
plt.ylabel('Residuals')
plt.title('Residual Plot: Engineered Model (Test Set)')
```

Write 3 to 5 sentences comparing this residual plot to your Phase 3 residual plot. Did the pattern improve? Are there still systematic errors? What does that tell you about what your model is still missing?

Part 5: Unified Comparison Table

Update your Phase 3 comparison table to include all models you have built across the full project. This table is the single most important artifact you will produce in Phase 4 because it is the evidence that drives your presentation.

Your table must have the following structure:

Model	MSE (train)	RMSE (train)	RMSE (test)	Improvement over Baseline
Constant Model	your value	your value	your value	N/A
Phase 3 SLR (retrained)	your value	your value	your value	your value
Engineered Feature Model	your value	your value	your value	your value

Write 3 to 5 sentences interpreting this table. Which model performed best? Did feature engineering provide meaningful improvement over your Phase 3 baseline? Is the gap between train RMSE and test RMSE consistent across models, and what does that tell you about overfitting?

Important: A model that barely improves over the constant model is not a failure. It is a finding. If your features have weak predictive signal, say so explicitly and explain why you think that is the case.

Part 6: Presentation Summary

This section is your presentation in written form. Complete each of the five subsections below. Your slides should be built directly from your answers here. If you write this section well, your presentation preparation is already 80 percent done.

6a: The Question

Write 3 to 5 sentences answering:

- What question were you trying to answer and why does it matter?
- Where did your data come from and what does it represent?
- What was your target variable and why is it worth predicting?

6b: What Your EDA Revealed

Write 3 to 5 sentences summarizing your Phase 2 findings. Your answer must:

- Name the most important pattern or relationship you found in the data
- Reference at least one specific visualization from Phase 2
- Explain how your EDA shaped your choice of feature in Phase 3

6c: What Your Models Did

Write 4 to 6 sentences summarizing your modeling arc. Your answer must:

- State your Phase 3 model equation explicitly
- Reference your unified comparison table from Part 5
- Explain what feature engineering you tried and whether it helped
- Compare your best model to the constant model baseline in plain English using the actual units of your target variable

6d: What Your Model Still Gets Wrong

Write 3 to 5 sentences being honest about your model's limitations. Your answer must:

- Reference your residual plot and describe any patterns you see
- Name at least one thing your model consistently over or under predicts
- Explain what that pattern tells you about what information is missing from your model

6e: What You Would Do Next

Write 3 to 5 sentences proposing concrete next steps. Your answer must:

- Name at least one additional feature you would add and explain why
- Name at least one modeling technique beyond linear regression that might help and explain your reasoning

- If you added supplementary data in Part 3d, explain what additional external data might further improve the model

A presentation that ends with 'my model worked well' with no acknowledgment of limitations will not receive full credit. The best presentations show that you understood not just what your model did, but what it cannot do and why.

Required Visualizations Summary

Your notebook must contain all of the following plots. Each must have a title, labeled axes, and a written interpretation. Plots without written interpretation will receive zero credit for that section.

Plot	Location	Min Length	Presentation Slide
Before/after scatter plot for transformed feature	Part 3a	2 to 3 sentences	Slide 3: EDA
Residual plot for engineered model on test set	Part 4	3 to 5 sentences	Slide 4: Model Results
Actual vs Predicted for best model on test set	Part 4	3 to 5 sentences	Slide 4: Model Results

Grading Criteria

Component	Points
Part 1: Written explanation of why Phase 3 evaluation was optimistic	10
Part 2: Train/test split applied correctly with comparison to Phase 3 RMSE	15
Part 3: At least two engineered features with justification and evidence	20
Part 4: Retrained model on engineered features with residual analysis	20
Part 5: Unified comparison table with interpretation	15
Part 6: Presentation summary with all five subsections completed	20
Total	100